

Exact Minimum and Maximum Aggregation for Finite First Passage

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Abstract

We derive the exact distributions of the minimum and maximum of every ordered pair of twenty frozen first-passage variables on uniform permutations of sixteen labels. Two-dimensional finite differences of the C90 joint survival arrays produce 400 joint laws, from which 13600 minimum/maximum coefficient cells and 13600 tail/CDF identities are certified. The pointwise sum identity and diagonal reductions are checked exactly. The result is finite combinatorics under `NO_BAD_EULER_OR_ROOT_NUMBER`.

1 Aggregation identities

Let $T_i \in \{0, \dots, 16\}$ and $S_{ij}(k, \ell) = \#\{T_i > k, T_j > \ell\}$. Finite Mobius inversion gives

$$N_{ij}(a, b) = S_{ij}(a - 1, b - 1) - S_{ij}(a, b - 1) - S_{ij}(a - 1, b) + S_{ij}(a, b).$$

Define $U_{ij} = \min(T_i, T_j)$ and $V_{ij} = \max(T_i, T_j)$. Then

$$\#\{U_{ij} > k\} = S_{ij}(k, k), \quad \#\{V_{ij} \leq k\} = 16! - S_i(k) - S_j(k) + S_{ij}(k, k).$$

Theorem 1. *For all 400 ordered pairs the derived minimum and maximum laws are nonnegative, normalized, transpose-compatible, and satisfy*

$$\mathbb{E}U_{ij} + \mathbb{E}V_{ij} = \mathbb{E}T_i + \mathbb{E}T_j.$$

On the diagonal, $U_{ii} = V_{ii} = T_i$ exactly.

Proof. The coefficient laws are pushforwards of the exact joint PMF. The two tail identities follow from the definitions and inclusion–exclusion. The expectation equality is the pointwise identity $\min(x, y) + \max(x, y) = x + y$. Diagonal and transpose assertions follow from $T_i = T_i$ and $N_{ij}(a, b) = N_{ji}(b, a)$. $\square$

certified object	count
ordered target pairs	400
minimum and maximum coefficient cells	13600
joint PMF cells independently checked	115600
minimum-tail / maximum-CDF identities	6800 / 6800
sum and diagonal identities	400 / 20

2 Scope

The package is a finite probability certificate. It makes no arithmetic/local data, Euler-factor, root-number, automorphy, full Burnside/table-of-marks, or Hilbert–Pólya operator claim.